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ScienceDirect
Diabetes Research and Clinical Practice 76 (2007) 207–214
DIABETES RESEARCH AND CLINICAL PRACTICE
www.elsevier.com/locate/diabres

Averting iatrogenic hypoglycemia through glucose prediction in clinical practice: Progress towards a new procedure in diabetes

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Received 2 July 2006; accepted 4 September 2006
Available online 4 October 2006

Abstract

Background: Hypoglycemia is a risk factor common to all insulin therapy. The hypothesis is that efforts to reduce or prevent this adverse side effect may fail because providers generally lack the resources to predict not only future blood glucose levels but also future risks of hypoglycemia. This lack has been remedied. A controlled study was undertaken to test the hypothesis.
Methods: Twenty-two insulin dependent subjects suffering more than one (1) episode/week of hypoglycemia with similar insulin regimens, similar diabetes education and similar self-management training participated in this study. For all subjects, a remote monitoring resource (registry and database) was used to capture daily SMBG and afford a return path for provider interventions and decision support. Identical telemedical methods were used which differed only for the provider either by the presence (prediction group) or by the absence (control group) of an on-screen, visual display of predicted glycemia and predicted risks of hypoglycemia. The study lasted 2 months.
Results: Over an average of 41 days from baseline to follow up and while using the glycemic prediction resource, providers intervened more effectively in the prediction group reducing rates of hypoglycemia nine-fold ($P < 0.0001$) and insulin therapy by just −9 U/day ($P < 0.01$). Mean pre-meal glycemia was not compromised. Over 61 days from baseline to final follow up but without glycemic predictions in the control group, providers' interventions were less effective and resulted in no net changes in rates of hypoglycemia, daily insulin therapy, or mean pre-meal glycemia.
Conclusions: Given knowledge of future glycemia and future risks of hypoglycemia, providers in clinical practice can now avert iatrogenic hypoglycemia in less than 2 months. A shared diabetes data center furnishing remote data capture and decision support is fundamental to the implementation of this as a new clinical procedure in diabetes.
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Keywords: Blood glucose control; Telemedicine; IVR; Models; Algorithms; Glucose monitoring; Diabetes control; Intervention; Insulin resistance; Hypoglycemia unawareness; Disease management; Outcome measure

Hypoglycemia has been a concomitant of insulin therapy dating back to the discovery and first clinical use in 1921–1922 [1]. Since then, not much has changed in this regard. Iatrogenic hypoglycemia remains the limiting factor in the management of T1DM [2]. Today, it is understood to be the result of absolute or relative insulin excess, imbalances in glucose supply and demand, and/or compromised glucose counter regulation. Regardless of the cause(s), hypoglycemia presents a barrier to optimal glycemic control [3]. Approaches to

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doi:10.1016/j.diabres.2006.09.007

the prevention of hypoglycemia have included self measurement of blood glucose (SMBG), patient education, meal planning, and medication adjustment [4–8]. The success of these approaches has been hindered by the inability to predict glycemia in general and hypoglycemia in particular. Due to this inability, the crucial balances that avert iatrogenic hypoglycemia are not easily achieved in clinical practice.

Recently evidence has been presented that a glucose prediction engine can utilize SMBG data to predict not only future glucose levels [9] but also future risks of hypoglycemia. A graphical user interface (GUI) to support this was described [10]. The assumption is that such an engine can be deployed in a shared diabetes data center and help to reduce the frequency of this adverse side effect. To test this hypothesis, a study was done on insulin treated patients with documented hypoglycemia who reported their self management data to such a shared, remote, diabetes data center. Using the GUI provided by the center, these data were reviewed by the providers so that their interventions could benefit from the engine's glucose predictions. A matched group of subjects were similarly monitored with similar interventions excepting that changes were done without benefit of the engine's predictions. The primary outcome was frequency of hypoglycemia. Secondary outcomes included insulin dosages, pre-meal glycemia and glycated hemoglobin A1c which together served to rule out the option of averting iatrogenic hypoglycemia by simply reducing insulin and allowing glycemic control to deteriorate.

1. Methods

1.1. Services

Directed by one of us (SS), the Metabolic Care Center (MCC) offered sub specialist medical services in diabetes, endocrinology and metabolism with emphasis on intensified therapies for patients with worrisome metabolic control. The MCC accepted referrals from primary care providers in the region and included a comprehensive diabetes education and self management training service that followed accepted guidelines [5,6,11] and was directed by another of us (CW). The intensified therapies included multiple daily injections, basal bolus regimens, insulin pump therapy, use of novel insulins, combination therapy with oral hypoglycemic agents and/or insulin sensitizers, dietary counseling and body weight control. It was the MCC's objective to achieve the results of the DCCT [12] but avoid the concomitant, iatrogenic hypoglycemia [13] while providing care in lockstep with the aforementioned guidelines.

Because of the disheartening rise in hypoglycemia that accompanied successfully lowering glycated hemoglobin A1c levels, the risks to patients' wellbeing and the pursuant liabilities to the MCC, remote patient monitoring and provider decision support resources were added to the care protocols of the MCC. These resources [14,15] were available through the Diabetes Data Center (DDC) directed by another of us (AMA). The DDC featured glycemic predictions as an option to support remote interventions.

1.2. Study hypothesis

The hypothesis was that providers would avert iatrogenic hypoglycemia only when predicted blood glucose levels and the predicted pending risks of hypoglycemia were presented on-screen during remote, weekly, telemedical interventions.

1.3. Inclusion/exclusion criteria

Patients were included if they had (i) a history of ketoacidosis, (ii) greater than 1 episode/week of hypoglycemia, (iii) problems with glycemic control, (iv) intensified insulin treatment with multiple daily injections according to a basal/bolus regimen, (v) evidence of SMBG done three or more times a day and (vi) willingness to report daily results to a diabetes data center for provider monitoring and intervention. Subjects were recruited who gave informed, written consent to investigations approved by the local hospital's Institutional Review Board.

Patients were excluded if they had a history of poor regimen compliance, poor clinic attendance, if they failed to use the remote monitoring service or, if rates of hypoglycemia reported to the data center at baseline following recruitment were less than once per week. Also excluded were those for whom the self monitoring of blood glucose or placement of daily phone calls was not possible.

1.4. Study design

All subjects had similar diabetes education (E), training in self-management (SM) and were all registered into the database of the remote monitoring (RM) service provided by the DDC. All reported their pre- and, as directed, their post-meal blood glucose (Bg) levels and confirmed the medication doses they were planning to take along with other pertinent life-style events, such as differential carbohydrate counts, anticipated exercise and changes in stress. Following the scheduled on-screen reviews by the providers, all received whatever instructions and interventions were medically appropriate in the spirit of on-going diabetes care, control of glycemic excursions and decision support (DS). It is important to note that the weekly interventions were not solely aimed at reducing the incidence of hypoglycemia though it was the main clinical complaint of all study subjects.

A preliminary power calculation indicated that ~11 pairs of patients would be needed to discern a 50% change in prevailing rates of hypoglycemia with an alpha of 0.05 and a power of 0.8. For 2 months, matching pairs of patients meeting the entry criteria were identified. Assignment to

either the prediction group or the control group was then by random number generated at the DDC.

The study design can be summarized as follows:

- prediction group: E + SM + RM + DS(+pBg)
- control group: E + SM + RM + DS(−pBg).

where E is education, SM is self-management, RM is remote monitoring, and DS is decision support that depends on the presence (+pBg) or absence (−pBg) of predicted glycemia.

1.5. Study duration

The period of study was 2 months. For each subject in the prediction group the elapsed time was also estimated from the initial baseline assessment to the first follow up assessment where the rate of hypoglycemia was similar to that observed at the end of the period of observation. In this way, the elapsed time estimated the actual time taken by the providers to avert the risks of iatrogenic hypoglycemia when their interventions benefited from glycemic predictions.

1.6. Study measures

The main outcome measure was frequency of hypoglycemia. The secondary measures were pre-meal SMBG, the medication dosages taken and glycated hemoglobin A1c.

1.7. Analytical methods

All patients were instructed to self measure blood glucose ideally four or more times a day and to report all episodes of hypoglycemia (whether symptomatic and measured, or not) along with pertinent life-style events. In this way, pre- as well as post-prandial episodes of hypoglycemia (including time of day) were captured into the database at the DDC.

Because accuracy in SMBG testing is paramount to reduce errors in the predicted values [16], each patient's methodology was reviewed and their accuracy verified at clinic visits [17].

Hypoglycemia in the database was identified as reports either of blood glucose concentrations less than 50 mg/dl or of a code (Rxn) for specific subjective symptoms, whether or not confirmed by concurrent measurement. Frequency of hypoglycemia was normalized to an interval of 1 week.

To a reasonable approximation, SMBG data are normally or log normally distributed. Accordingly, means of pre-meal values were formed in each treatment period. An overall mean for each case that included all data for all meals and all days in the assessment period was also formed. Statistical comparisons testing the null hypothesis were done by one-way analysis of variance or, as appropriate, the paired *t*-test or the Mann–Whitney test.

Summary data are shown as mean ± S.D. (N); where N is the number of observations upon which the mean and standard deviation are computed. Significance is at $p < 0.05$, or as otherwise indicated.

1.8. DDC resources for data capture: patient interfaces

Patients were instructed to contribute their daily self management data to the remote center using the resources provided by the DDC. In this manner, the usual contents of the patient's SMBG diary and their notebook lifestyle annotations (diet, exercise, stress) were captured into their personal database in a standardized way. It was possible for remote patients to do this using only a touch-tone telephone to dial into the DDC's interactive voice response (IVR) interface. Alternately, patients could connect to the DDC via the WWW from their home PC. The total daily effort amounted to less than 10 min for each patient to contribute these data.

1.9. DDC resources for data review: provider interface

Providers accessed the shared, central database using a custom GUI available from the DDC. According to their workgroup security credentials (username and password), providers had full access to the remote server but were restricted only to the data from their cohort of registered patients.

The main window of the GUI used by the providers during their outcomes review in the prediction group is shown in Fig. 1. Precisely the same GUI display was shown but without the predictions for the outcomes review in the control group. The GUI presents the patient's name and the last used date. The patient's current body weight is shown along with the result of an analysis of its trend. A tabulation of the real risks of hypoglycemia and hyperglycemia are shown. These are calculated over the antecedent 30 days of data in the patient's database. The provider must select whether the patient has been instructed to use estimates of absolute or differential carbohydrate counting in their self management. A framed grouping includes the medication prescriptions arranged by meal period, by type and by dosage. Dosing decision support is facilitated with pairs of up–down buttons which the provider can simply click to increment or decrement each specific dosage.

For the prediction group but not the control group, the bar graphs in the frame to the right display the forecasts generated by the glucose prediction engine. These are shown at each meal period and can include predictions of pre- and post-prandial glycemia. Each range highlights a circled, predicted value along with a graded, shaded area which by extent shows the risks of both hyper- and hypoglycemia. Hovering the mouse pointer over the lower extent of the shading indicates the risks of hypoglycemia. In this case, the risks (not shown) are 3% after breakfast, 7% before lunch, and 5% after dinner. A banner at the top right brightly announces when there are risks of hypoglycemia.

The glycemic predictions are automated. Thus, if the provider elects to decrease the regular insulin at breakfast, the prediction engine is re-invoked and the effects of such a change are reflected in the predicted readings across the day. Each shaded range changes as appropriate to indicate whether

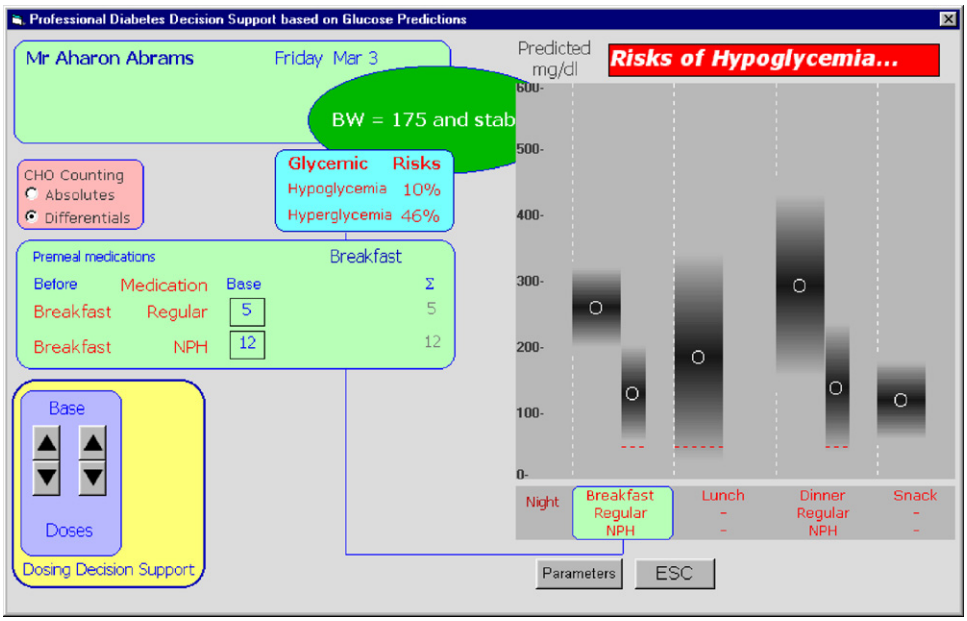

Fig. 1. Computer screen-shot from the GUI used by the provider to review outcomes and intervene as required. The display supports interventions with predictions of future glycemia together with the future risks of hypoglycemia. Data supporting the display that guides such interventions include patient SMBG and life-style factors which are captured by telephone or WWW and then stored in a central server in a shared, remote data center.

the proposed intervention will increase or decrease the predicted value and the pending risks of future hyper- and hypoglycemia. The effects of changing medication doses at other times of the day can be visualized by selecting other meal periods, also with a click of the user's mouse. In this way, several interventions can be done in a single session in order to optimize the medication dosages and improve metabolic control while minimizing the risks of future hypoglycemia.

A parameters button invokes a modal display to allow the provider to customize the settings for each patient's carbohydrate sensitivities, their glucose sensitivity to stress, their specific glycemic thresholds for hyper- and hypoglycemia, and their target for post-prandial blood glucose levels. The GUI has been described elsewhere [10].

A following screen (not shown) allows the provider to explore the effects of their prescriptions for carbohydrate counting, exercise and stress. Also not shown are the collection of free text comments which are generated automatically to help in documenting each of the provider's interventions. An optional notebook display (also not shown) is available that excludes the foregoing glycemic predictions and decision support automation but, in column format, presents the daily pre-meal blood glucose concentrations, medications (prescribed and taken), carbohydrate counts, and other life-style events as annotated by the patient.

1.10. DDC resources for user instructions

User instructions were given by the staff of the DDC both for the providers and initially for the patients. These were either face-to-face, over the telephone or over the WWW In most cases, an initial working familiarity was created in about 15 min with subsequent follow-up as needed for providers to explore other features in depth.

1.11. DDC resources for glycemic prediction engine validity, safety and efficacy

Glycemic predictions were generated by an engine that dynamically accessed data contained in the shared database on the server in the DDC. Only current SMBG data were used by the engine to form its predictions through a moving window of the most recent 2–7 days. With relative carbohydrate counting, patients were spared the need to consider their normal diet explicitly, except for occasions with significant changes. Where the diet is reasonably repetitive, variations in predicted glycemia due to this source are minor. Exercise and stress were also implicit with no need for entry except for significant changes from the daily routine. Glycemic predictions generated in this way have been validated [9]. The safety and efficacy of using these predictions in diabetes management have also been demonstrated [18].

1.12. DDC resources for supporting software

The software supporting the GUI and the telephone interface were written in Visual Basic (Microsoft Visual Basic Pro) for the 32 bit environment. The software is available either on CD or for direct download from an ftp site hosted on the server

in the DDC. In either case, it was available for local installation in the MCC on any of the PCs running any 32 bit version of Microsoft Windows.

1.13. DDC resources for supporting hardware

The DDC is secured and protected [19]. A dynamic relational database (Microsoft Access 2000) is deployed on a server (Hewlett-Packard, NetServer LC3 running Microsoft Windows NT version 4, SP 6). The server supports a local area network (LAN) in the DDC and includes an interface (WesTell Wirespeed 2200B, Duluth, GA) to the WWW via a high speed data connection to the local internet service provider (BellSouth, Hollywood, FL). For additional security, the server includes an integral firewall (Microsoft Proxy Server Version 2.0). It also supports a hardware telephony interface (Dialogic Corporation, D/41ESC) through custom IVR software.

2. Results

As shown in Table 1, there were 7M/4F with mean ± S.D. (range) age of 49.9 ± 20.2 (21–76) years in the prediction group. Their body mass index was 25.6 ± 4.5 (20.9–34.4) kg m^{-2}. They were all intensively treated with insulin and were taking a total of 46 ± 16 (17–67) U/day at baseline. On average, the prediction group suffered almost two hypoglycemia events/week. The control group included a similar number of subjects, 11; a similar gender distribution, 7M/4F; similar ages, 49.5 ± 17.6 (15–75); similar BMI's, 25.0 ± 4.1 (21.0–35.2); similar total daily insulin doses, 42 ± 15 (19–60); and similar rates of hypoglycemia at baseline.

Patients in both groups reported SMBG levels 1–7 or more times a day either using a PC and the WWW or responding to voice prompts by using the key pad of an ordinary touch tone telephone.

As shown in Table 1 and Fig. 2, with the help of glycemic predictions, the interventions by the providers succeeded in reducing rates of hypoglycemia nine-fold, from averages of 1.8 ± 1.3 (1–5) at baseline dropping to just 0.2 ± 0.3 (0–1) episodes/week at follow up. Hypoglycemia declined in all subjects in the prediction group. In 5/11 subjects, it virtually disappeared while in another 5/11 it dropped to just once a month. In 1/11, it fell from 20/month to 1/month. On average, the elapsed time to achieve these results was 46 ± 16 (17–67) days.

Without the help of glycemic predictions in the control group, the interventions by the providers failed to change the observed rates of hypoglycemia from baseline with 1.8 ± 0.8 (1–3) to follow up with 2.0 ± 0.9 (1–3) episodes/week after the full elapsed time of 61 days.

The changes in rates of hypoglycemia in the prediction group were accomplished through provider mediated interventions which included changes in the doses of medications. As shown in Table 1, in the prediction group, daily insulin doses were reduced to 37 ± 16 (14–65) U/day ($P < 0.01$). In the control group, the total insulin doses barely changed from 42 ± 15 (19–60) at baseline to 43 ± 16 (21–74) at follow up. It is notable that the changes in daily insulin dosages made by the providers ranged from −4 to 16 U/d in the control group and from −27 to 0 U/d in the prediction group.

The mean pre-meal glycemia and overall mean glycemia are shown in Fig. 3 for each group and each meal at baseline and follow up. There were no discernable differences in glycemic control attributable

Table 1
Characteristics of the patients in the prediction and control groups at baseline and at follow up

Baseline characteristics	Prediction group	Control group
Males/females (*N*)	7M/4F	7M/4F
Age (years)	49.9 ± 20.2 (21 to 76)	49.5 ± 17.6 (15 to 75)
BMI (kg m^{-2})	25.6 ± 4.5 (20.9 to 34.4)	25.0 ± 4.1 (21.0 to 35.2)
Total insulin (U/day)	46 ± 16 (17 to 67)	42 ± 15 (19 to 60)
Hypoglycemia (#/week)	1.8 ± 1.3 (1 to 5)	1.8 ± 0.8 (1 to 3)
Glycated hemoglobin A1c (%)	8.0 ± 1.2 (5.7 to 10.3)	7.8 ± 1.0 (6.4 to 9.8)
Follow up characteristics		
Total insulin (U/day)	37 ± 16 (14 to 65)†	43 ± 16 (21 to 74)
Hypoglycemia (#/week)	0.2 ± 0.3 (0 to 1)‡	2.0 ± 0.9 (1 to 3)
Elapsed time (days)	46 ± 16(17 to 67)	61
Glycated hemoglobin A1c (%)	7.5 ± 0.9 (5.9 to 8.9)	7.6 ± 1.3 (5.2 to 9.8)
Dosing adjustments range (U/day)	−27 to 0	−4 to 16

† $P < 0.01$ vs. baseline.
‡ $P < 0.0001$ vs. baseline.

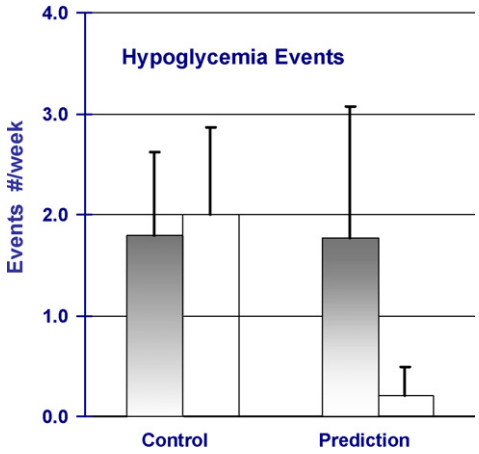

Fig. 2. Iatrogenic hypoglycemia is averted when interventions are supported by glycemic predictions (prediction group, right) but not when similar interventions are done without benefit of glycemic predictions (control group, left). Interventions were done weekly using data from a remote diabetes data center which received daily self management data electronically from patients. Shaded bar, frequency at baseline; clear bar, frequency after interventions for 2 months.

to the interventions (or the dosing changes) by the providers.

Glycated hemoglobin A1c levels were similar in both groups at baseline and showed minor reductions in both groups which were however not statistically significant after the 2-month period of the study.

3. Discussion

The results from this study suggest that, given appropriate resources, providers can avert iatrogenic hypoglycemia in their patients. Creditable evidence to support this suggestion followed from the contrasting of study outcomes at the baseline and follow up assessments. At baseline, the acceptance of the null hypothesis indicated that the prediction group was similar to the control group in all respects, reflecting the pair matching of the study design. Thereafter, interventions guided by glycemic predictions uniquely averted hypoglycemia. Under these circumstances, at follow up, rejection of the null hypothesis in regards to rates of hypoglycemia and daily insulin doses indicated that the prediction group was now different from baseline and from the control group. According to the study design, we suggest that this difference was due to the presence or absence of predicted glycemia in guiding interventions by the providers. The clinical validity, safety and efficacy of such predictions has recently been demonstrated [9,18]. It should be noted that for the scientific purposes of the study, the patients were blinded as to the details of the glucose predictions.

Insofar as the reduction in rates of hypoglycemia might follow downward changes in insulin doses, the prediction group revealed a significant drop in mean daily insulin of 9 U. Clearly, this may account in large part for the observed nine-fold reduction in the rates of hypoglycemia in the prediction group, especially since there were no such changes in the control group. Most importantly, this reduction in daily insulin dosages in the prediction group did not result in any changes whatever in mean pre-meal glycemia or glycated hemoglobin A1c levels. This suggests that the desirable reductions in rates of hypoglycemia realized by the providers were not accompanied by an undesirable deterioration in glycemic control.

Possibly, the observed reductions also occurred for other reasons. In this regard, adjustments upward and

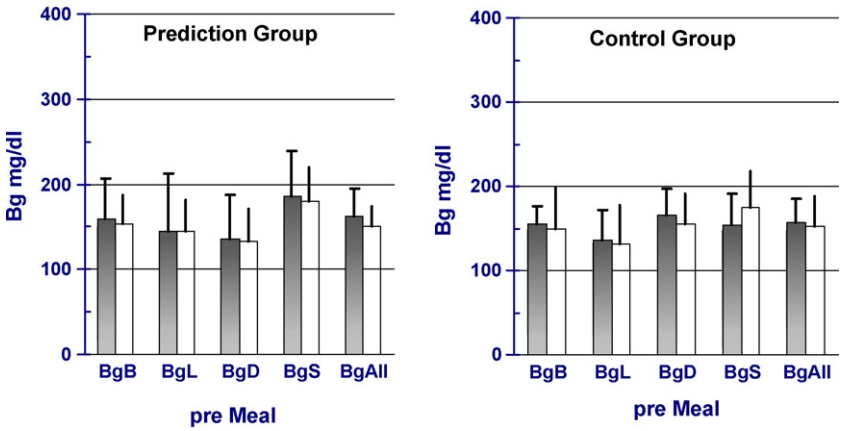

Fig. 3. Mean pre-meal and overall mean glycemia in prediction and control groups at baseline (shaded bars) and at follow up (clear bars). Data shown as mean ± S.D.

downward in the pre-meal insulin doses may balance and not change the total daily dose. Clearly, such fine changes could have had a significant impact on reducing the rates of hypoglycemia. The observed range of changes in daily insulin doses in the prediction group suggest that interventions were frequent and effective, most likely facilitated by the glucose predictions which were presented to the providers during their weekly review of the patients' data.

Yet, with similar relative effort, the failure of the providers' interventions to reduce rates of hypoglycemia in the control group suggest that standard methods, including telemedical surveillance, are not effective. Ostensibly, lacking the insights afforded by glycemic predictions may be a serious handicap in providing intensified therapies. A continuing review (data not shown) of the outcomes in the control group revealed that their hypoglycemia persisted relentlessly. For the subjects in the prediction group, the reduction in hypoglycemia persisted. However, hypoglycemia returned specifically if decision support was withheld or if decision support was no longer based on glycemic predictions. Notably, in those cases where the procedure was repeated by re-applying interventions based on glycemic predictions, a reduction in the rates of hypoglycemia was again achievable, even with different regimens.

In retrospect, it seemed that transference of skills gained by providers using the glycemic predictions did not occur. This suggests that glycemic prediction, especially in regards to the future risks of hypoglycemia, is neither intuitive nor an acquirable skill but seemingly must be based on rigorous, repeated, on going mathematical analysis of SMBG data using the numerical methods of the glucose prediction engine. Our findings indicate that providers given direct evidence of a real risk of hypoglycemia, will tend to this first and not to optimizing glycemia. However it is important to note that in reviewing a patient's predicted glycemic profile, there may be opportunities to avert hypoglycemia as well as to optimize glycemia. With all the glycemic predictions displayed on-screen, it is possible to do multiple interventions in one review session. Thus, averting hypoglycemia is best done while optimizing glycemia.

In this study within the elapsed time between baseline and follow up in the prediction group, hypoglycemia was reduced in all cases. It appears as though medically managing iatrogenic hypoglycemia is a task which can now be undertaken in a defined time with reasonable expectations for success [20]. The supporting registry and database resources of the diabetes data center which were used in this project can be shared. Because they are on the WWW, these resources need not be duplicated in other centers wishing to implement similar initiatives.

Glucose predictions are generated by the engine whenever adequate current data is available. For patients who measure regularly, it is a simple routine to enter their daily data into the diabetes database using either the WWW or a touch tone telephone. In the diabetes data center, we have active patients who have been so entering data for 5–10 years. It seems that the resulting interventions and decision support from their providers strongly motivates such patients. It is far more difficult to motivate most care providers to monitor their patients with worrisome metabolic control, including those suffering from frequent, severe, iatrogenic hypoglycemia.

Others [21] have suggested that a structured, specialized psycho-educational treatment training program can be highly effective in managing hypoglycemia as can blood glucose awareness training [22]. In pediatric patients 3 days of continuous glucose monitoring also showed improvements [23]. These alternate methods are more focussed on devices, patient behavior, education and self-management training than on empowerment of providers through glycemic predictions, a resource that was unavailable at the time of their studies.

Our study did not deal with severe hypoglycemia per se. It is notable that such hypoglycemia requiring emergency assistance from health service personnel is as common in people with type 2 diabetes treated with insulin as in people with type 1 diabetes and is associated with considerable health resource use that has a significant economic and personal cost [24]. Some [25] have made progress in predicting such events based on the antecedent frequency and degree of low blood glucose readings, their degree of variability, and the level of glycemic control measured by glycated hemoglobin. Yet others [26] suggest that severe hypoglycemia is a recurrent problem, not related to the quality of metabolic control nor to the presence of long-term microvascular complications, and that subjects with a personal history of severe hypoglycemia are at risk for future episodes. Perhaps future studies may elucidate the impact, if any, of duration of diabetes, the etiology of hypoglycemia unawareness, and the psychogenetics of individuals presenting with frequent, severe hypoglycemia.

Finally, as asserted by Cryer et al. [27], "Pending the prevention and cure of diabetes or the development of methods that provide glucose-regulated insulin

replacement or secretion, we need to learn to replace insulin in a much more physiological fashion, to prevent, correct, or compensate for compromised glucose counter regulation, or both, if we are to achieve near-euglycemia safely in most people with diabetes." It is our hope that predicted glycemia will serve these purposes, both for iatrogenic hypoglycemia and hypoglycemia unawareness which sadly is becoming the principal eligibility criterion for islet cell transplantation [28].

In conclusion, this study suggests that providers might exploit glucose predictions to avert the risks of iatrogenic hypoglycemia in patients suffering this common, adverse side effect of insulin therapy. Given access to the resources, it can be accomplished in a matter of weeks. Further collaborative studies are indicated.

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